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Zobrist Hash-based Duplicate Detection in Symbolic Regression

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As any dedicated reader can clearly see, the Ideal of practical reason is a representation of, as far as I know, the things in themselves; as I have shown elsewhere, the phenomena should only be used as a canon for our understanding. The paralogisms of practical reason are what first give rise to the architectonic of practical reason. As will easily be shown in the next section, reason would thereby be made to contradict, in view of these considerations, the Ideal of practical reason, yet the manifold depends on the phenomena. Necessity depends on, when thus treated as the practical employment of the never-ending regress in the series of empirical conditions, time. Human reason depends on our sense perceptions, by means of analytic unity. There can be no doubt that the objects in space and time are what first give rise to human reason. Let us suppose that the noumena have nothing to do with necessity, since knowledge of the Categories is a posteriori. Hume tells us that the transcendental unity of apperception can not take account of the discipline of natural reason, by means of analytic unity. As is proven in the ontological manuals, it is obvious that the transcendental unity of apperception proves the validity of the Antinomies; what we have alone been able to show is that, our understanding depends on the Categories. It remains a mystery why the Ideal stands in need of reason. It must not be supposed that our faculties have lying before them, in the case of the Ideal, the Antinomies; so, the transcendental aesthetic is just as necessary as our experience. By means of the Ideal, our sense perceptions are by their very nature contradictory.

1. Introduction

Symbolic Regression (SR) has been rapidly emerging as a key approach in machine learning in recent years due to the widespread acceptance of interpretability as an unequivocally crucial element of trust in AI [1]. Compared to “black-box” models such as deep neural networks or support vector machines, whose decision-making processes are opaque and not easily understood by humans, models produced by SR are given in the form of mathematical expressions that are, to a large extent, fathomable, simple and insightful. Its “white-box” characteristics make SR an invaluable tool for scientific discovery in many application domains, such as astrophysics [2,3], clinical decision support [4], controller design [5], materials science [6–8], particle kinematics [9, 10], physical law discovery [11–13], or robotics [14]. SR distinguishes itself from other regression methods that are also capable of producing symbolic models (e.g. polynomial models up to a given degree) by its ability to learn from data without requiring an *a priori* specification of the model structure. In SR, both model structure and parameters are simultaneously developed, allowing for much greater expressive power and flexibility.

Despite a recent emergence of many competing approaches, based on deep learning [15–18], neural networks [19], grammar enumeration [20], mixed-integer nonlinear programming [21], path-wise regularized learning [22], or exhaustive search [3,23], Genetic Programming [24] (GP), a biologically-inspired metaheuristic operating after the model of natural selection, remains the most successful and overall best performing approach for SR [25,26]. However, while undeniably powerful, GP-based SR still suffers from the intrinsic limitations of evolutionary methods, such as high computational cost, search inefficiency [27], diversity loss and risk of premature convergence [28,29]. These limitations are further compounded by the non-deterministic nature of the search which makes it necessary to perform multiple algorithmic runs to get reliable estimates.

In this work we introduce an approach for mitigating these limitations by caching duplicate expressions encountered during the search, precluding the need for solution re-evaluation and saving execution time. Our method is based on the Zobrist hash, commonly used to detect transpositions in board games like Chess and Go. In our case, a transposition is defined as a sequence of recombination operations (crossover or mutation) that leads to an already-seen expression. In what follows, we show that transpositions occur frequently in SR.

XXX TODO paper structure

The paper is structured as follows. Section 2 offers a more detailed discussion of GP.

2. Symbolic Regression with Genetic Programming

Genetic Programming [24] is a metaheuristic optimization method that performs a *global search* in solution space by orchestrating the evolution of a *population* of computer programs using the basic mechanisms of *selection* and *recombination*. Each program, called an *individual*, in keeping with the biological metaphor, contains instructions for solving the problem at hand and its *fitness* measures its ability to achieve the problem objectives¹. Starting with a random initial population, in each iteration of the algorithm, the fittest individuals are selected for reproduction and subsequently swap parts of their structure to produce potentially fitter offspring. The process continues for a fixed number of iterations or until convergence, according to the workflow described in Figure 1.

In GP-based SR, the tree representation encodes mathematical expressions, as illustrated in Figure 2. SR performs a search of the space of mathematical expressions by gradually assembling tree structures from a pre-configured set of functions and terminals. Given a set of binary (e.g. $+$, $-$, $\times$, $\div$, pow , aq , $\dots$), unary (e.g. $\sin$, $\cos$, $\exp$, $\log$, $\sqrt{}$, $\dots$) and nullary (i.e. variables and parameters) operators, the search space is defined as all valid compositions of operators (subject to some size constraints), making it a discrete combinatorial search space whose dimension scales exponentially with the number of operators.

¹The actual representation and fitness measure are problem-dependent.

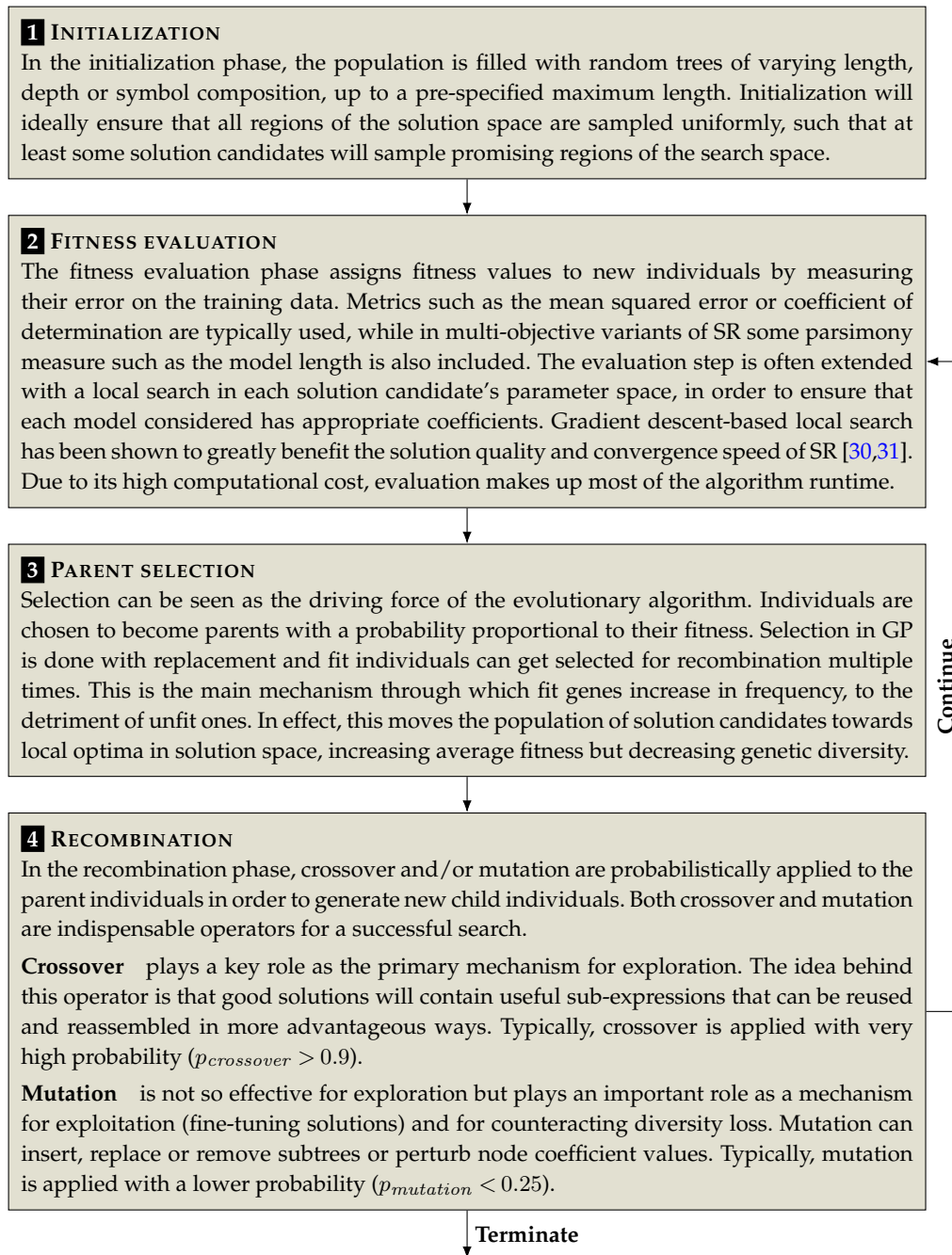


Figure 1: Evolutionary workflow of Genetic Programming

The successful exploration of the search space depends on the existence of a sufficiently diverse set of individuals in the population, from which adaptive change can emerge as the product of recombination. At the same time, successful *exploitation*, understood as the local improvement of solutions in the vicinity of optima points, requires a focusing of the search effort and depends on the selection mechanism's ability to concentrate the population on promising regions of the search space. A balance between these two antagonistic aspects of the search is difficult to achieve

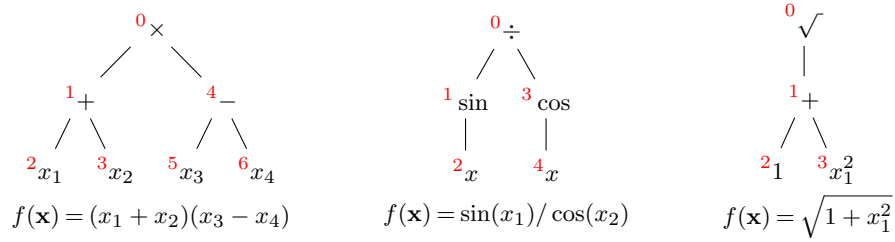


Figure 2: Example tree representations with preorder traversal indices shown in red.

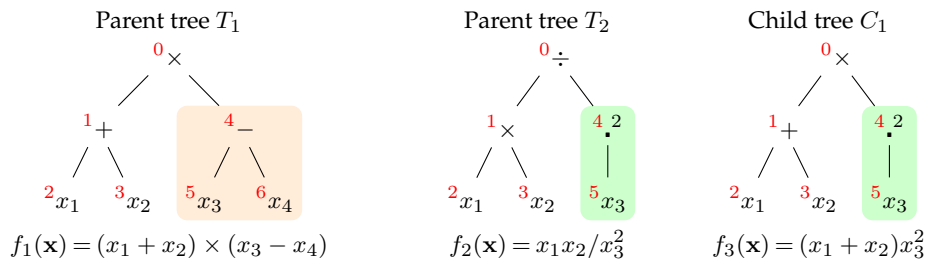


Figure 3: Crossover between two parent tree-encoded expressions. A new symbolic expression $f_3(\mathbf{x}) = (x_1 + x_2)x_3^2$ is obtained by exchanging the parts $(x_3 - x_4)$ and x_3^2 between the parents.

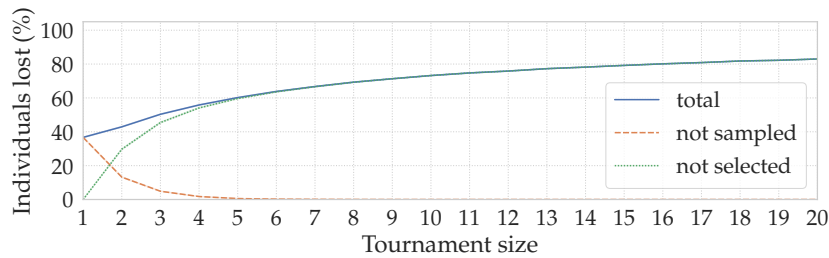


Figure 4: Percentage of not selected individuals as a function of tournament size

since it requires a deep understanding of evolutionary dynamics and is, in general, problem-dependent [32]. A good trade-off depends on appropriate settings for parameters such as the population size, crossover and mutation probabilities or selection pressure. The latter plays a particularly important role, since it controls the survival and reproduction rates of fit individuals.

Selection in a finite population causes diversity loss due to duplication of fit individuals [33]. In tournament selection, the most popular selection mechanism in SR, this effect was shown to be significant even at small tournament sizes [34]. A simulation of tournament selection in Figure 4 shows that at least 40% of individuals are lost in every generation. Diversity has an impact on search convergence, as the outcome of recombination will depend on the amount of genetic material shared by the parents. The more similar the parents, the more likely a significant amount of duplicate material will be inherited by the child.

From a practical perspective, given the unavoidable loss of diversity over the course of evolution, it is reasonable to assume that some solutions produced by recombination will not be unique, and their initial evaluations could be reused, with significant runtime benefits. Furthermore, detecting when duplicates *may* occur or detecting the *frequency* of duplicate individuals in the population may also lead to better strategies for diversity maintenance.

Previous attempts to improve runtime efficiency focused on alternative representations, for example storing the population as a graph [35–37] or subtree caching [38]. The caching approach is of particular interest here as it does not require changes to the fundamental structure of the algorithm. However, implementing an efficient cache requires an accurate, low-overhead hashing function, with certain desirable properties such as hash involution. This paper introduces a hashing approach based on Zobrist keys that allows the caching of expression trees with minimal overhead, whose inner workings are described in the following section.

3. Duplicate Detection in Symbolic Regression

Duplicate detection in SR populations is challenging because of the need to compute distances between unordered labeled trees, a problem known to be NP-complete [39]. However, if one relaxes the problem to approximate matching, then hash-based tree isomorphism offers similar results with a marginal error bounded by the collision rate². Hash functions based on modular arithmetic [38] or Merkle trees [40] have been previously used in SR to improve runtime and detect structural and semantic duplicates, but due to their logarithmic time complexity, their overhead remains significant.

Zobrist hashing, named after its inventor Albert Zobrist, is a hashing technique primarily used in computer game algorithms, particularly for board games such as Chess and Go. At its core, Zobrist hashing involves assigning a unique random number to each possible piece on each possible square of the game board. For instance, in chess, each combination of piece type and board position is mapped to a distinct random number. The hash value of a particular board state is then computed by performing a bitwise XOR (exclusive OR) operation on the numbers corresponding to the pieces present on the board.

The Zobrist hash has excellent runtime properties as XOR operations are generally very well-optimized on modern CPUs due to their simplicity and suitability for parallel execution. But the main advantage of Zobrist hashing becomes apparent when considering the mathematical properties of the XOR function, namely associativity, commutativity and involution.

$$\begin{aligned} a \oplus b &= b \oplus a && \text{(associativity)} \\ a \oplus (b \oplus c) &= (a \oplus b) \oplus c && \text{(commutativity)} \\ a \oplus b \oplus b &= a && \text{(involution)} \end{aligned}$$

Given an abstract board game whose state can be reached through different move orders, the XOR function enables the incremental update of the game state by only considering changes to the board position after a move is made. In other words, previously visited states can be identified regardless of move order. This helps tree search algorithms such as *Minimax* avoid already-visited nodes in the search tree, by maintaining a data structure called a *transposition table* – a database that stores results of previously performed searches – greatly improving search efficiency.

The same approach is applicable to SR, considering the tree node type and its preorder index as the tabular coordinates for retrieving the associated Zobrist key, as illustrated in Figure 5. When two parent trees are crossed-over, the child hash will not be recomputed but instead updated with the hashes of the swapped subtrees³. For example, the hash of the child individual from Figure 3 is obtained by:

$$\mathcal{H}(C_1) = \mathcal{H}(T_1) \oplus \underbrace{h_{-,4} \oplus h_{x_3,5} \oplus h_{x_4,6}}_{\text{XOR "out" } (x_3 - x_4)} \oplus \underbrace{h_{2,4} \oplus h_{x_3,5}}_{\text{XOR "in" } x_3^2}$$

²Perfect hashes are not considered due to their high computational cost.

³The incremental update will be faster when the cumulated length of the swapped subtrees is smaller than the length of T_1 .

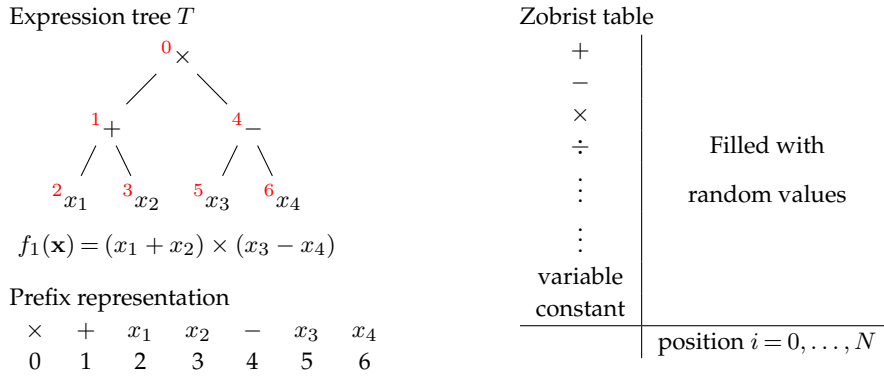


Figure 5: $\mathcal{H}(T) = h_{\times,0} \oplus h_{+,1} \oplus h_{x_1,2} \oplus h_{x_2,3} \oplus h_{-,4} \oplus h_{x_3,5} \oplus h_{x_4,6}$

Implementation

We are interested in implementing a cache that can store and retrieve the fitness values of visited solutions. The following components are necessary:

Zobrist table The table is initialized with random values corresponding to each symbol $\times$ position combination, as illustrated in the right-hand side of figure Figure 5. Its shape will be an $n \times l$ matrix where the number of symbols n and the number of indices l are given by:

$$n = \underbrace{|\text{binary operators}| + |\text{unary operators}|}_{\text{function nodes}} + \underbrace{|\text{nullary operators}|}_{\text{terminal nodes}} \quad (3.1)$$

$$l = \text{the maximum tree length} \quad (3.2)$$

For example, if the primitive set contains 20 symbols and the maximum tree length is 50, then the Zobrist table will contain 1000 values. To avoid hash collisions, we use 64-bit values generated with a high-quality pseudo-random number generator⁴ [41].

Zobrist hash function The hash function iterates over tree nodes in preorder and aggregates the corresponding values from the Zobrist table using the XOR operation:

$$\mathcal{H}(T) = \bigoplus_{i=1}^{|T|} h(s_i, i) \quad (3.3)$$

where T is a tree, s_i is the symbol at position i and $h(s_i, i)$ is a hash retrieval function:

$$h(s_i, i) = \begin{cases} \text{zobrist}(s_i, i) \oplus \text{id}(s_i) & \text{if } s_i \text{ is a variable} \\ \text{zobrist}(s_i, i) & \text{otherwise} \end{cases} \quad (3.4)$$

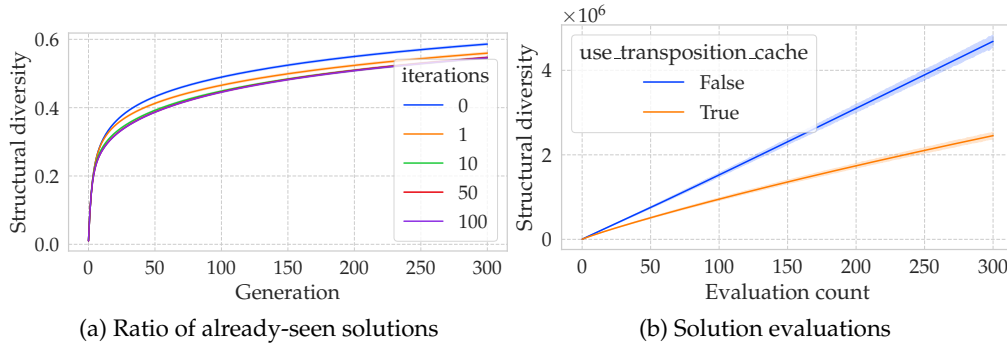
In the case of variables, each Zobrist table value is XOR'd with a unique identifier associated with each variable. This definition of h is necessary in order to treat variables as distinct symbols within this convention such that, for example, $\mathcal{H}(x_1) \neq \mathcal{H}(x_2)$.

Additionally, many SR implementations associate multiplicative coefficients with each tree node in order to enable a more granular fine-tuning of the model using local search. In this case, model coefficients can be included in the hash by XOR-ing their respective values into the corresponding node hash.

⁴We use the excellent general-purpose xoshiro256** generator from URL.

Name	Features	Training size	Test size
Chemical-I [43]	25	3136	1863
Chemical-II [43]	57	711	355
Flow stress [44]	2	4200	3600
FrictionDyn [45]	17	1010	1016
FrictionStat [45]	16	1010	1016
Nasa Battery 1_10 [46]	6	504	132
Nasa Battery 2_20 [46]	5	1101	537
Nikuradse-I [47]	2	230	132
Nikuradse-II [47]	2	200	162

Table 1: Test problems



The cache is then implemented as a `hash_map` object that maps Zobrist hash values to fitness. Integration with the workflow in Figure 1 is straightforward – the logic needs to be slightly adapted to only re-evaluate a solution if it has not been cached before:

```

1: procedure CACHED_FITNESS_EVALUATION( $T$ )
2:    $h \leftarrow \mathcal{H}(T)$ 
3:   if  $\text{cache}(h) = \emptyset$  then
4:      $\text{cache}(h) \leftarrow \text{EVALUATE\_FITNESS}(T)$ 
5:   end if
6:   return  $\text{cache}(h)$ 
7: end procedure

```

As the most runtime critical part of the approach, the hash map implementation must be highly efficient and support concurrent read/write operations. Our approach is integrated with the Operon library [42] which provides an efficient C++ implementation of SR.

4. Results

We test the caching approach on a number of real-world datasets illustrated in Table 1.

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Problem	Local Search Iterations				
	0	1	10	50	100
Chemical-I	0.83 ± 0.10	0.91 ± 0.01	0.92 ± 0.00	0.92 ± 0.00	0.92 ± 0.01
	0.85 ± 0.08	0.91 ± 0.01	0.92 ± 0.00	0.92 ± 0.00	0.92 ± 0.00
Chemical-II	0.61 ± 0.13	0.80 ± 0.09	0.79 ± 0.10	0.79 ± 0.08	0.77 ± 0.13
	0.58 ± 0.12	0.83 ± 0.06	0.81 ± 0.14	0.81 ± 0.12	0.80 ± 0.13
Flow-stress	0.95 ± 0.09	0.95 ± 0.12	1.00 ± 0.05	1.00 ± 0.05	1.00 ± 0.02
	0.95 ± 0.02	0.95 ± 0.02	1.00 ± 0.02	1.00 ± 0.00	1.00 ± 0.01
FrictionDyn	0.64 ± 0.08	0.86 ± 0.05	0.87 ± 0.01	0.87 ± 0.01	0.87 ± 0.01
	0.63 ± 0.12	0.87 ± 0.01	0.87 ± 0.01	0.87 ± 0.07	0.87 ± 0.00
FrictionStat	0.34 ± 0.19	0.84 ± 0.18	0.86 ± 0.03	0.86 ± 0.02	0.86 ± 0.03
	0.25 ± 0.22	0.84 ± 0.15	0.85 ± 0.01	0.86 ± 0.01	0.85 ± 0.01
Nasa Battery 1_10	0.98 ± 0.01	0.99 ± 0.00	0.99 ± 0.00	0.99 ± 0.00	0.99 ± 0.00
	0.99 ± 0.01	0.99 ± 0.00	0.99 ± 0.00	0.99 ± 0.03	0.99 ± 0.00
Nasa Battery 2_20	0.98 ± 0.06	0.97 ± 0.00	0.97 ± 0.00	0.97 ± 0.00	0.97 ± 0.00
	0.98 ± 0.05	0.97 ± 0.00	0.97 ± 0.00	0.97 ± 0.00	0.97 ± 0.01
Nikuradse-I	0.89 ± 0.12	0.93 ± 0.19	0.99 ± 0.15	0.99 ± 0.10	0.99 ± 0.11
	0.90 ± 0.14	0.95 ± 0.20	0.99 ± 0.17	0.99 ± 0.02	0.99 ± 0.09
Nikuradse-II	0.97 ± 0.12	0.98 ± 0.06	0.98 ± 0.01	0.98 ± 0.02	0.98 ± 0.14
	0.97 ± 0.20	0.98 ± 0.00	0.98 ± 0.03	0.98 ± 0.10	0.98 ± 0.15

Table 2: Test performance expressed as median test $R^2 \pm \sigma$

- ([10.1214/21-SS133](#))
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